NOGA AHARONY

PhD Candidate in Systems Biology, Columbia University

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Computational biology PhD candidate developing evaluation frameworks and applications for genome language models. My work asks what these models learn, where they fail, and how they can be used in microbial genomics and biosurveillance, spanning genomic context, pathway-level reasoning, metagenomic binning, and detection across evolutionary distance.

EDUCATION

In progress	**Doctor of Philosophy, Systems Biology**, Columbia University Thesis: Systematic Evaluation Frameworks for Foundational Genome Language Models Development Advisor: Mohammed AlQuraishi, PhD
2021	**Master of Science, Biology**, Technion - Israel Institute of Technology Thesis: Rapid Gene Content Alteration in Recurring Infections Advisor: Roy Kishony, PhD
2019	**Bachelor of Science, Honours Neuroscience**, McGill University First Class Honours; Dean's Honours List; Dean's Multidisciplinary Research List

CURRENT FELLOWSHIPS AND AFFILIATIONS

Aug–Oct 2026	**Pivotal Research Frontier Biodefense Fellow** Developing reference-free methods to detect engineered DNA in metagenomic sequencing data.
Sep–Dec 2026	**ML Alignment & Theory Scholars (MATS), Autumn 2026** Biosecurity track, mentored by Gary Abel (Fourth Eon Bio), focused on technical research at the intersection of AI and biosecurity, including function-based DNA synthesis screening.

RESEARCH EXPERIENCE

2022-Present	**Mohammed AlQuraishi Lab**, Columbia University, Program of Mathematical Genomics Developing evaluation frameworks and applications for biological language models, especially genome language models, across genomic context use, pathway-level reasoning, wastewater pathogen monitoring, and protein-protein interaction prediction. Co-developing GeneCtx for dissecting genomic context use and WAVEBench for evaluating genomic foundation models under realistic metagenomic distribution shifts.
2021-2023	**Tal Korem Lab**, Columbia University, Program of Mathematical Genomics Created a model to detect small RNAs in bacterial genomes.
2022	**Itsik Pe'er Lab**, Columbia University, Department of Computer Science Developed a neural embedding model in hyperbolic spaces to better predict small proteins.
2019-2021	**Roy Kishony Lab**, Technion - Israel Institute of Technology, Department of Biology Constructed a computational pipeline to identify mobile genetic elements in recurring infections. Designed experiments unraveling eco-evolutionary dynamics of microbial communities.
2018-2019	**Amine Kamen Lab**, McGill University, Department of Bioengineering Designed and manufactured adeno-associated viruses carrying Cas9 for improved CAR-T cell engineering.

2018	**Center for Health Security**, Johns Hopkins University, School of Public Health Surveyed DNA synthesis providers' sequence- and customer-screening practices; this work informed the Health Security governance paper *Strengthening Security for Gene Synthesis*. Studied safety, innovation, and community norms in the DIY biology community.
2017-2018	**Jonathan Kimmelman Lab**, McGill University, Biomedical Ethics Unit Applied machine learning to analyze clinical trial design factors that influence forecasts of outcomes; contributed to a meta-analysis.
2017	**Edward Ruthazer Lab**, McGill University, Montreal Neurological Institute Electroporated and imaged tadpoles to study structural plasticity in the developing visual system.
2016-2017	**Rafael Najmanovich Lab**, University of Montreal, Department of Pharmacology and Physiology Developed software characterizing protein interfaces based on surrounding force fields. Modeled biophysical features of *C. difficile* germination protease to identify candidate inhibiting ligands.

PUBLICATIONS AND MANUSCRIPTS

2026	Le Lievre T*, **Aharony N***, Wang S, Adams E, AlQuraishi M. GeneCtx: An Evaluation Framework for Dissecting Genomic Context Use in Genome Language Models. Submitted to NeurIPS 2026 Evaluations & Datasets track.
2023	Liang C, Wagstaff J, **Aharony N**, Schmit V, Manheim D. Managing the Transition to Widespread Metagenomic Monitoring: Policy Considerations for Future Biosurveillance. *Health Security*, 21(1), 34-45.
2022	Myers A, Utpala S, Talbar S, Sanborn S, Shewmake C, Donnat C, Mathe J, Lupo U, Sonthalia R, Cui X, Szwagier T, Pignet A, Bergsson A, Hauberg S, Nielson D, Sommer S, Klindt D, Hermansen E, Vaupel M, Dunn B, Xiong J, **Aharony N**, Pe'er I, Ambellan F, Hanik M, Nava-Yazdani E, von Tycowicz C, Miolane N. ICLR 2022 Challenge for Computational Geometry & Topology: Design and Results. arXiv.
2021	Milman O*, Yelin I*, **Aharony N**, Katz R, Herzel E, Ben-Tov A, Kuint J, Gazit S, Chodick G, Patalon T, Kishony R. Community-level evidence for SARS-CoV-2 vaccine protection of unvaccinated individuals. *Nature Medicine*.
2020	Yelin I*, **Aharony N***, Shaer-Tamar E*, Argoetti A*, Messer E, Berenbaum D, Shafran E, Kuzil A, Gandali N, Hashimshony T, Mandel-Gutfreund Y, Halberthal M, Geffen Y, Szwarcwort-Cohen M, Kishony R. Evaluation of COVID-19 RT-qPCR test in multi-sample pools. *Clinical Infectious Diseases*.
2019	Moco PD, **Aharony N**, Kamen A. Adeno-Associated Viral Vectors for Homology-Directed Generation of CAR-T cells. *Biotechnology Journal*.

* Equal contribution.

POSTERS AND TALKS

2026	**Aharony N**, AlQuraishi M. Evaluating Genome Language Model Embeddings as Transferable Priors for Metagenomic Binning. Accepted poster, CIBB 2026. Rome, Italy, Sep 2–4, 2026.
2026	**Aharony N**, Wang S, Neiswanger W. Benchmarking Genome Language Models for Novel Virus Detection in Wastewater Across Evolutionary Distance. Selected oral presentation, ASM BIG 2026. Washington, DC, Oct 11–14, 2026; talk Oct 13.
2026	**Le Lievre T*,** Aharony N*, **AlQuraishi M. PRIME-Bench: A Benchmark for Pathway-Level Reasoning in Genome Language Models. Poster, ISMB 2026.**
2026	**Aharony N**. Who Holds the Reins in Agent-Assisted Systems Biology? Accepted poster and extended abstract, NE AI Agents Day 2026.

2026	**Aharony N**, AlQuraishi M. Evaluating Genome Language Models for Metagenomic Binning. Accepted poster, Pioneering Israeli Science Conference / ScienceAbroad 2026.
2026	**Aharony N**. Rapid Talks and Panel: 2024 Innovative Tools & Solutions Award Recipients. Flash talk and panelist, PRI.
2025	**Aharony N**. Genome Language Models Reading the Future of Pandemics in Wastewater. Flash talk, PRI Innovation Showcase.
2022	**Aharony N**. Computational Methods for Pandemic Pathogen Detection. Flash talk, Singer Symposium.
2021	**Aharony N**, Kishony R. Rapid Alteration in Genetic Content Upon Recurring Infection. Poster, EMBL Symposium: New Approaches and Concepts in Microbiology.
2020	**Aharony N**. Careers in Biosecurity. Flash talk, EAGx Virtual.
2018	**Aharony N**, Gronvall G. How Secure is the Gene Synthesis Industry? Poster, Biological Weapons Convention: Meeting of Experts.

AWARDS AND FUNDING

2026	Pivotal Research Frontier Biodefense Fellowship.
2026	ML Alignment & Theory Scholars (MATS) Program, Autumn 2026.
2024	NYC-PRI Innovative Tools and Solutions to Address Public Health Emergencies.
2022	EA Funds PhD Support Scholarship.
2021	Open Philanthropy Early-Career Funding.
2020	Miriam and Aaron Gutwirth Memorial Fellowship for Excellence in Research.
2019	Leonard and Diane Sherman Interdisciplinary Graduate School Fellowship.
2018	Open Philanthropy Early-Career Funding for Global Biological Risks.
2017, 2018	McGill University Faculty of Science Scholarship.
2017	CIHR Undergraduate Research Award in Computational Biology.
2016	NSERC Undergraduate Student Research Award; FRQNT Top-Up to NSERC Undergraduate Student Research Award.

TEACHING, MENTORSHIP, AND SCIENTIFIC SERVICE

2016-2018	**Teaching Assistant**, McGill University Created problem sets, held office hours, and led review sessions for biochemistry, molecular biology, neuroscience, and introductory physics courses.
2018-2023	**Research and career mentorship** Mentored students and early-career professionals interested in biosecurity and computational biology projects through Effective Thesis Project and 80,000 Hours referrals.
2019	**Volunteer Consultant**, Battelle Memorial Institute / ThreatSEQ Consulted on a ThreatSEQ joint initiative with the FBI assessing whether gene synthesis companies would detect orders containing genes from the Select Agents list.

SKILLS

Computational	Python, R, MATLAB, C/C++, Java; biological language model evaluation; sequence modeling; metagenomic analysis; hyperbolic embeddings; genomic pipeline development.
Experimental	Microbial evolution experiments; AAV vector production; CRISPR/Cas9 delivery; molecular biology workflows.
Languages	Hebrew (native), English (near-native), French (novice).